

Wei-Jhih Huang

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Research Interests

AI-Native Radio Access Networks, Learning-Based Resource Orchestration, Deep Learning for Cross-Domain Applications

Education

Bachelor, Communication Engineering, National Central University
GPA: 3.94/4.3, Expected Graduation: July 2026

Aug 2022 – Present
Taoyuan, Taiwan

Research Experience

Wireless Mobile Network Lab, National Taiwan University
Research Assistant, Advisor: **Prof. Hung-Yu Wei**

Apr 2026 – Present
Taipei, Taiwan

- Developed an AI-native orchestration framework for LLM deployment across distributed edge networks, resolving end-to-end latency by strategically balancing communication overhead with computational resource availability.
- Designed predictive resource management algorithms within a 6G AI-RAN architecture, utilizing cross-layer telemetry to anticipate network dynamics and enable proactive data staging for seamless service continuity in highly mobile scenarios.

Information Processing and Communications Lab, National Central University
Undergraduate Research Student, Advisor: **Prof. Chih-Wei Huang**

Feb 2024 – Present
Taoyuan, Taiwan

- Proposed a self-aware Meta-RL framework for 6G radio resource management with a Feature-wise Attention Gate enabling KNN-based OOD detection of traffic shifts; achieved **0.790** AUROC and **1.51 ms** latency on 100 3GPP 5QI-compliant scenarios.
- Built a trajectory-embedding mechanism for scenario distribution-aware DRL fine-tuning, reducing unnecessary DRL retraining by **90%** and improving user quality of service by **11%** in cross-layer resource management scenarios.

Electronic Design Automation Lab, The Chinese University of Hong Kong
Summer Visiting Research Student, Advisor: **Prof. Bei Yu**

Jul – Aug 2025
Hong Kong

- Enhanced *NUA-Timer++* for graph-based hierarchical circuit timing prediction under Non-Uniform Input Arrival Times, achieving a **1.96x** improvement in consistency and reducing average error to **0.42x** compared to SOTA.
- Collaborated with a postdoctoral researcher on Transformer architecture design and loss function refinement for *NUA-Timer++*; independently investigated a critical-path-based timing constraint extension as an auxiliary signal for logic synthesis.

Publications

- **Wei-Jhih Huang**, Jin-Min Yang, Ning Ma, Sheng-Liang Wu, Chih-Wei Huang. "Deviation-Aware Trajectory Embedding for Fine-Tuning Triggering in DRL-Based Wireless Network Resource Management," *In Proceedings of the IEEE Wireless Communications and Networking Conference (WCNC)*, 2026.
- Ziyi Wang, **Wei-Jhih Huang**, Fangzhou Liu, Tsung-Yi Ho, David Z. Pan, Bei Yu. "NUA-Timer++: Timing Prediction for Hierarchical RTL under Non-Uniform Input Arrival Times," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, Under Review.
- Ching-Wei Lin, **Wei-Jhih Huang**, Sheng-Liang Wu, Chih-Wei Huang, Phone Lin, Shun-Ren Yang. "PACI-Net: Physics-Aware Distillation for Explainable Radio Resource Management," *IEEE Global Communications Conference (GLOBECOM)*, Under Review.

Honors & Awards

- **Gold Award for Service Learning**, National Central University, 2026
- **Outstanding Student Scholarship**, National Central University, 2026
- **Outstanding College Youth Award**, School Level, Taiwan, 2026
- **Undergraduate Research Fellowship**, National Science and Technology Council, Taiwan, 2025–2026
- **Honorable Mention**, Undergraduate Project Competition, Department of Communication Engineering, NCU, 2025
- **Merit Award, Individual Service-Learning Excellence Scholarship**, National Central University, 2025
- **High Distinction, Public Service Scholarship**, National Central University (as President, NCU Rural Service Club), 2024

Teaching & Work Experience

Chunghwa Telecom <i>AI Software Development Intern</i> - Spearheaded the full-stack development of an intelligent project management system, spanning UI/UX design and n8n workflow backend orchestration, driven by RAG-based agentic models to enhance operational efficiency by 33% .	Sep – Dec 2025 Taipei, Taiwan
National Central University <i>Teaching Assistant, Courses: Natural Language Processing and Enterprise Application of Pattern Recognition</i> - Assisted in lecture preparation and grading for NLP topics for 60 students and image recognition topics for 60 students, analyzing the underlying AI technologies of real-world company products to support students in final project research.	Feb – Jun 2025
Industrial Technology Research Institute <i>AI Software Summer Intern</i> - Built and deployed an LLM-based service chatbot at <i>Taipei Songshan Sports Center</i> , integrating Azure OpenAI with Linux backend services to provide 24/7 automated customer support, reducing customer queue time by 50% and improving service efficiency.	Jun – Sep 2024 Hsinchu, Taiwan